

A General Purpose Computer-Assisted Clustering Methodology

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Joint work with Justin Grimmer (Harvard $\rightsquigarrow$ Stanford)

A Method for Computer Assisted Conceptualization

- Conceptualization through **Classification**: “one of the most central and generic of all our conceptual exercises. . . . the foundation not only for conceptualization, language, and speech, but also for mathematics, statistics, and data analysis. . . . Without classification, there could be no advanced conceptualization, reasoning, language, data analysis or, for that matter, social science research.” (Bailey, 1994).

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- (We focus on clustering texts; methods apply more broadly)

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- Fully automated algorithms can help, but which ones?

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- No surprise: everyone's tried cluster analysis; very few are satisfied

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- **The Question: How to organize all those clusterings?**

Our Idea: Meaning Through Geography

Set of clusterings

Set of clusterings $\approx$
A list of unconnected addresses

[illegible]

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A list of unconnected addresses

wide at **SuperPages.com**

	195	Car	C
Carterge New England Inc 26 Allen Ln Ipswich 01938... 978 356-9960 Carterge Lydia 38 Sweet Rd 02131... 617 323-7639	Carter F 24 Hickox Rd 02131... 617 327-1105 Faye & Ricky 107 Columbia Ave 02136... 617 437-7331 Francis S 134 Temple W Ave 02132... 617 323-6781 Franklin & Anne 293 Mt Auburn Cam 02136... 617 354-0798 Fred 42 Haverhill Ave 02136... 617 524-3078 Fred 36 Haverhill Ave 02136... 617 524-3078 G & E 8 Haverhill Ave 02136... 617 436-8906 G T 27 Fyfield Ave 02136... 617 623-7121 Gayle 25 Franklin Ave 02136... 617 623-0322 Geo S 115 Main Hill Rd Jan 02136... 617 522-3215 George 125 Nashua Rd 02136... 617 367-9548 Carter Holiday Associate 107 S Street Box 02111... 617 456-1689 Carter Harry F 30 Baring Rd W Jan 02132... 617 325-5465	Carter Nellie E 323 Main St Box 02135... 617 267-6483 Nicholas S F 115 Randolph Ave 02136... 617 698-5307 Nick T 21 Fairfield Box 02116... 617 267-5222 Nick & Debbi 156 Vermont Rd Newton 02459... 617 527-0480 Norman G 38 Chickadee Dr 02125... 617 822-1203 P E 501 E South St Box 02037... 617 268-4213 P L 44 Matthews Box 02131... 617 427-9170 P R 91 Boyer Jan 02136... 617 968-8692 Paul & Constance 114 Adams Ave W Jan 02130... 617 325-2036 Paul E 501 E South St Box 02037... 617 268-4546 Paul M 27 Union St 02135... 617 787-2115 Carter Pile Driving Inc 17 Beaver Ct Franklin 02102... 617 393-3782 Carter Prudence 40 Franklin Waterbury 02172... 617 393-3782 Prudence 40 Franklin Waterbury 02172... 617 393-3782 Reginald 106 Broadview Cambridge 02215... 617 541-2843 Renee & Andrew 10 Walnut Box 02108... 617 720-3765 Carter Rice Dowd Baker Dennis Publishing 163 Main Wilmington 01897 Toll Free 800-1 & 2 then... 800 638-1671 Cost Rec Industrial Prod 613 Main Wilmington Toll Free 800-1 & 2 then... 800 619-7447 Toll Free 800-1 & 2 then... 800 648-7447 Headquarters 613 Main Wilmington 01897 Call... 978 988-7447 Ingalls Crane 363 Main Wilmington 01897 Call... 978 988-7447 Carter Richard 2075 Carver Ave Brighton 02137... 617 987-0836 Richard A 9748 Vernon Ave 02106... 617 566-7293 Carter Richard A MD 170 0th Ave Box 02136... 617 267-0710 Carter Richard K 133 Merwin St 02135... 617 268-9468 Robert L 175 Rockdale Ave Can 02104... 617 864-1535 Roger 130 St Paul Ave 02131... 617 424-6148 Roy M Concord 02138... 617 491-6115 Royce 185 Salisbury Cha 02129... 617 241-9418	



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		195	Car	C
37 566-1282	Cartage New England Inc	28 Allen Ln Ipswich 01908	978 356-9960	
81 447-4101	Cartagena Lydia	38 Sweet Box 02131	617 323-7639	
90 257-9961	Cartagena Aveth	1 Fairview Rd 02139	617 442-9780	
37 566-1282	Cartagena Aveth	1 Fairview Rd 02139	617 442-9780	
37 566-1282	Cartagena Aveth	1 Fairview Rd 02139	617 442-9780	
361-0380	Carte Nicholas	38 Avenue Box 02131	617 695-6996	
37 566-4548	Carte Nicholas	38 Avenue Box 02131	617 695-6996	
37 628-8248	Carten Thos J Sr & Claire	1 Laverne Rd 02131	617 338-0219	
37 445-5116	Carten Thos J Sr & Claire	1 Laverne Rd 02131	617 338-0219	
37 822-9992	Carte A Inc	50 Thompson Ln 02136	617 696-6919	
37 442-5712	Carte A Inc	50 Thompson Ln 02136	617 696-6919	
37 569-2698	Carte A Inc	50 Thompson Ln 02136	617 696-6919	
37 667-5190	Carte A Inc	50 Thompson Ln 02136	617 696-6919	
37 569-1417	Carte A Inc	50 Thompson Ln 02136	617 696-6919	
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37 296-4725	Carte A Inc	50 Thompson Ln 02136	617 696-6919	
37 542-1521	Carte A Inc	50 Thompson Ln 02136	617 696-6919	
37 364-5232	Carte A Inc	50 Thompson Ln 02136	617 696-6919	
37 541-5649	Carte A Inc	50 Thompson Ln 02136	617 696-6919	
37 739-2662	Carte A Inc	50 Thompson Ln 02136	617 696-6919	
37 879-0030	Carte A Inc	50 Thompson Ln 02136	617 696-6919	
37 541-3948	Carte A Inc	50 Thompson Ln 02136	617 696-6919	
37 836-1511	Carte A Inc	50 Thompson Ln 02136	617 696-6919	
37 569-4119	Carte A Inc	50 Thompson Ln 02136	617 696-6919	
37 800-4782	Carte A Inc	50 Thompson Ln 02136	617 696-6919	



$\approx$ We develop a (conceptual) geography of clusterings

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- 7 **↪ Millions of clusterings, easily comprehended** (takes about 10-15 minutes to choose a clustering with insight)

You choose one (or more), based on insight, discovery, useful information,...



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- Meila (2007): derives same metric using different axioms (lattice theory)

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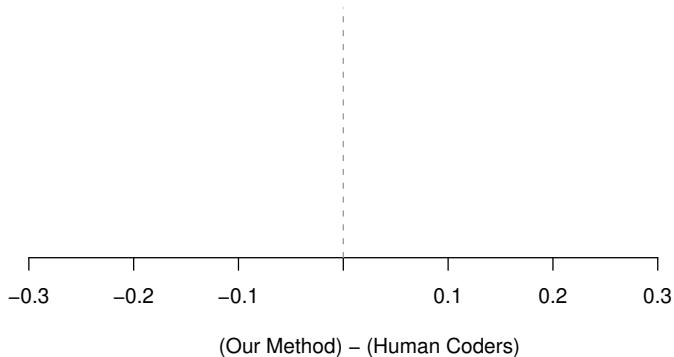
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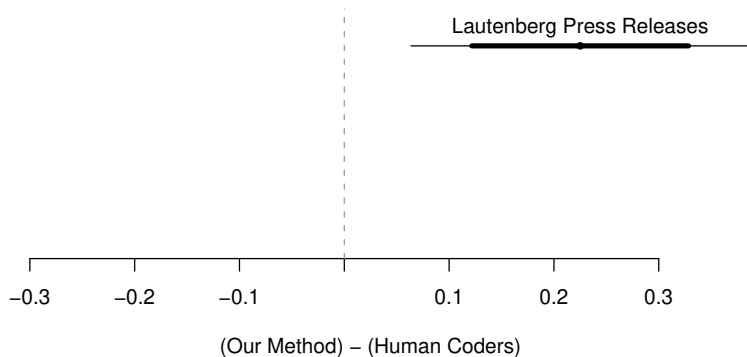
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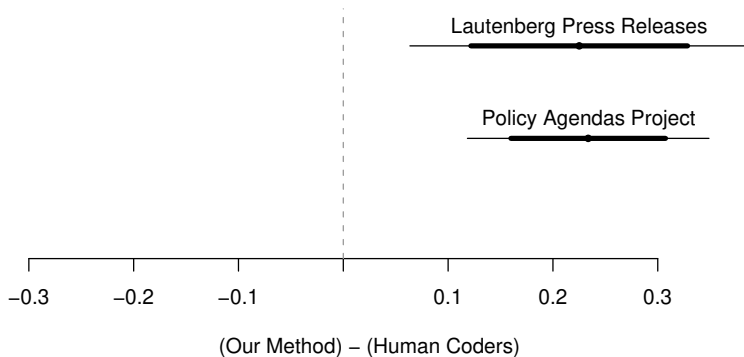


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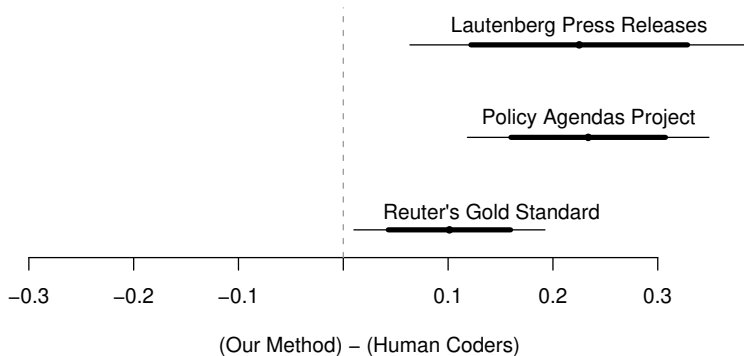
Lautenberg: 200 Senate Press Releases (appropriations, economy, education, tax, veterans, ...)

Evaluation 1: Cluster Quality



Policy Agendas: 213 quasi-sentences from Bush's State of the Union (agriculture, banking & commerce, civil rights/liberties, defense, ...)

Evaluation 1: Cluster Quality



Reuter's: financial news (trade, earnings, copper, gold, coffee, . . .); "gold standard" for supervised learning studies

Evaluation 2: More Informative Discoveries

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“Immigration”:

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“Genetic testing”:

Our Method 1 $\rightarrow$ {Our Method 2, K-Means 1, K-means 2} $\rightarrow$ Dir Proc. 1 $\rightarrow$ Dir Proc. 2

Evaluation 3: What Do Members of Congress Do?

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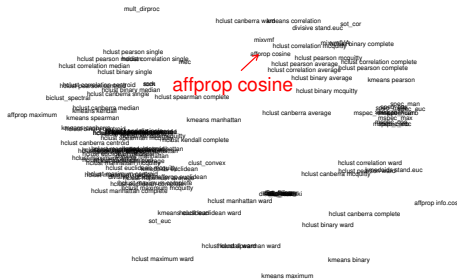
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- Apply our method

Gary King (Harvard IQSS)

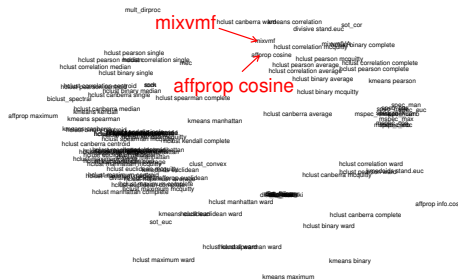


Example Discovery



Red point: a **clustering** by Affinity Propagation-Cosine (Dueck and Frey 2007)

Example Discovery



Red point: a **clustering** by
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Close to:

Mixture of von Mises-Fisher distributions (Banerjee et. al. 2005)

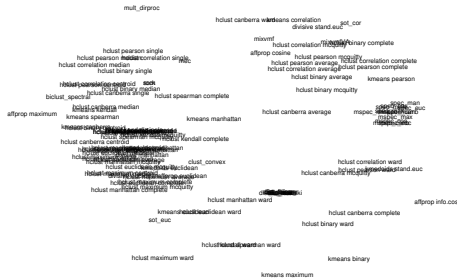


Example Discovery

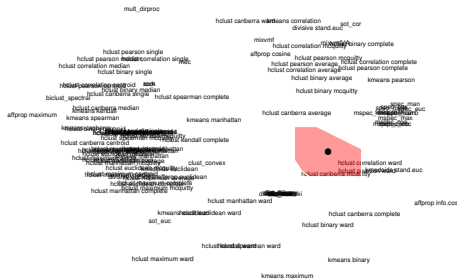


Space between methods:
local cluster ensemble

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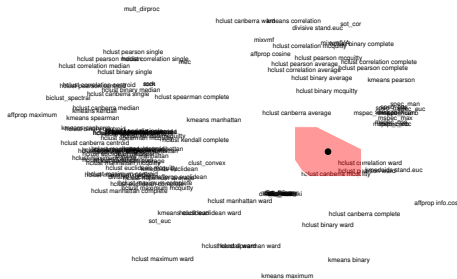


Example Discovery



Mixture:

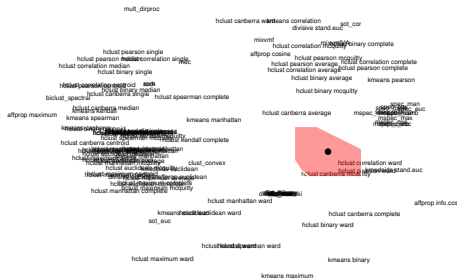
Example Discovery



Mixture:

0.39 Hclust-Canberra-McQuitty

Example Discovery

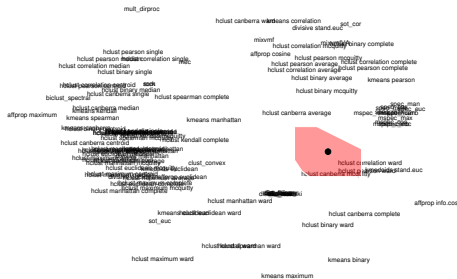


Mixture:

0.39 Hclust-Canberra-McQuitty

0.30 Spectral clustering
Random Walk
(Metrics 1-6)

Example Discovery



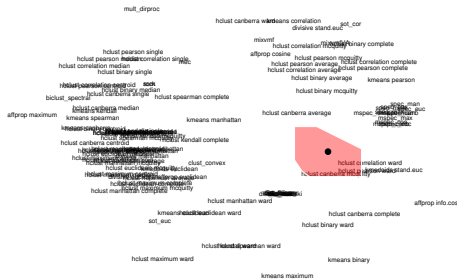
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Random Walk
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Example Discovery



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0.30 Spectral clustering
Random Walk
(Metrics 1-6)

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0.05 Kmediods-Cosine

Example Discovery



Mixture:

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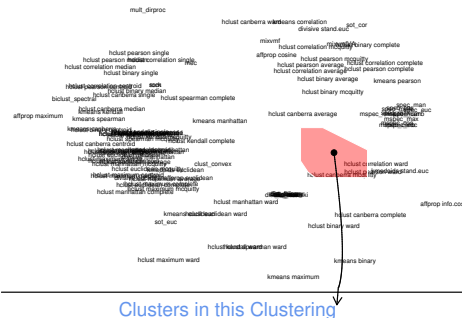
0.04 Spectral clustering

Symmetric
(Metrics 1-6)



Mayhew

Example Discovery

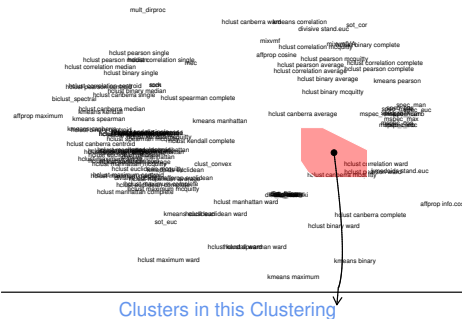


Credit Claiming
Pork

Credit Claiming, Pork:
 “Sens. Frank R. Lautenberg (D-NJ) and Robert Menendez (D-NJ) announced that the U.S. Department of Commerce has awarded a \$100,000 grant to the South Jersey Economic Development District”

Mayhew

Example Discovery



Credit Claiming, Legislation:

"As the Senate begins its recess, Senator Frank Lautenberg today pointed to a string of victories in Congress on his legislative agenda during this work period"

Credit Claiming Pork

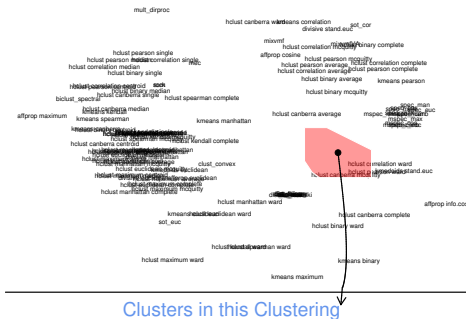
Mayhew

Credit Claiming Legislation

Gary King (Harvard IQSS)

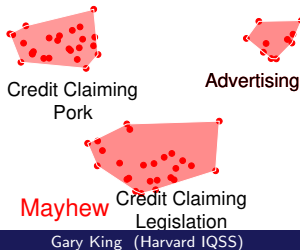
Quantitative Discovery

Example Discovery

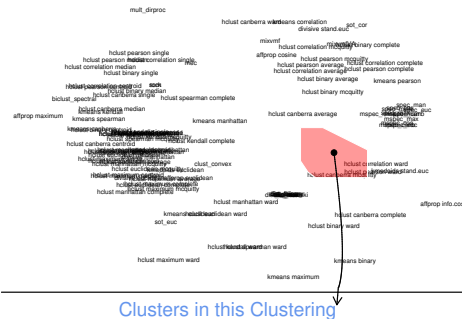


Advertising:

“Senate Adopts
Lautenberg/Menendez Resolution
Honoring Spelling Bee Champion
from New Jersey”

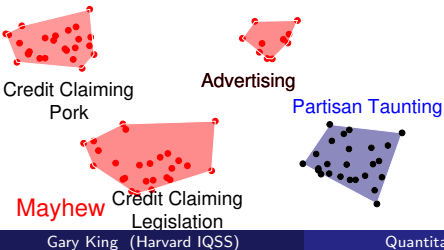


Example Discovery: Partisan Taunting

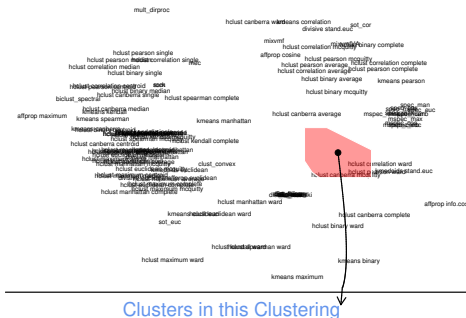


Partisan Taunting:

“Republicans Selling Out Nation on Chemical Plant Security”



Example Discovery: Partisan Taunting



Partisan Taunting:

“Senator Lautenberg’s amendment would change the name of . . . the Republican bill . . . to ‘More Tax Breaks for the Rich and More Debt for Our Grandchildren Deficit Expansion Reconciliation Act of 2006’”



Credit Claiming
Pork

Advertising

Partisan Taunting

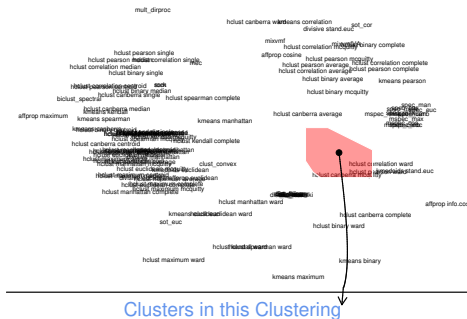
Mayhew

Credit Claiming Legislation

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Quantitative Discovery

Example Discovery: Partisan Taunting



Definition: Explicit, public, and negative attacks on another political party or its members

Clusters in this Clustering



Credit Claiming Pork



Advertising



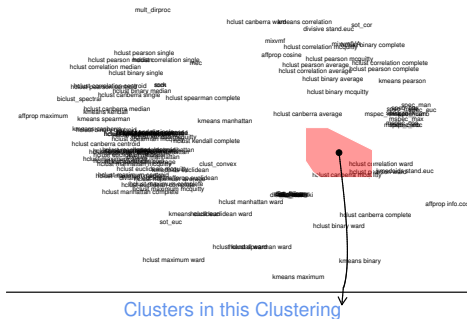
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Credit Claiming Legislation



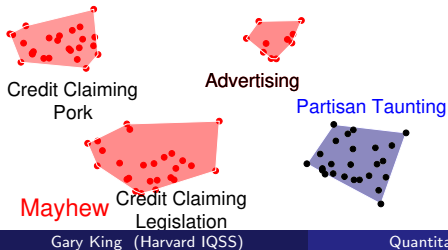
Partisan Taunting

Example Discovery: Partisan Taunting



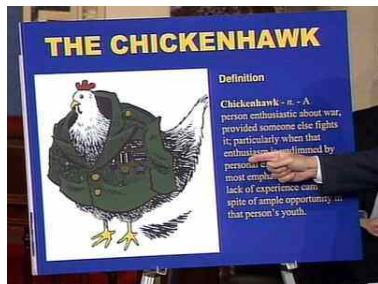
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Taunting ruins deliberation



In Sample Illustration of Partisan Taunting

Taunting ruins deliberation

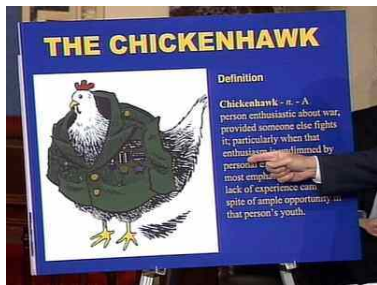


Sen. Lautenberg
on Senate Floor
4/29/04

- "Senator Lautenberg Blasts Republicans as 'Chicken Hawks' " [Government Oversight]

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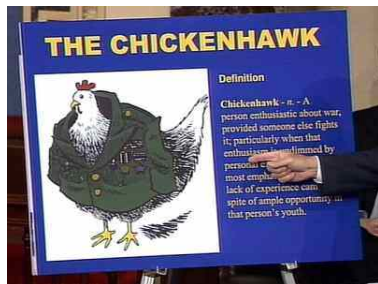


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- "Every day the House Republicans dragged this out was a day that made our communities less safe." [Homeland Security]

Out of Sample Confirmation of Partisan Taunting

- Discovered using 200 press releases; 1 senator.

Out of Sample Confirmation of Partisan Taunting

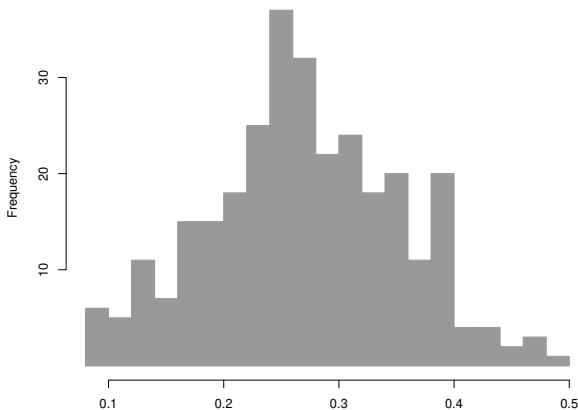
- Discovered using 200 press releases; 1 senator.
- Confirmed using 64,033 press releases; 301 senator-years.

Out of Sample Confirmation of Partisan Taunting

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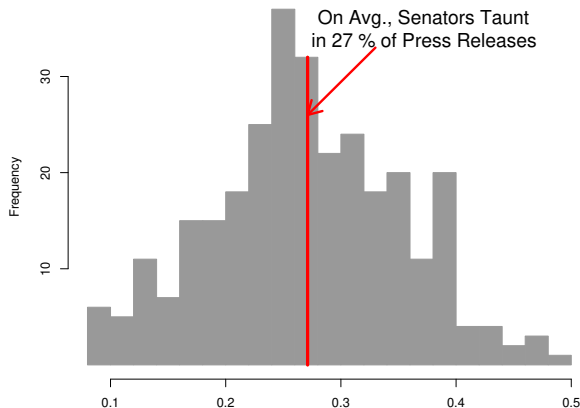
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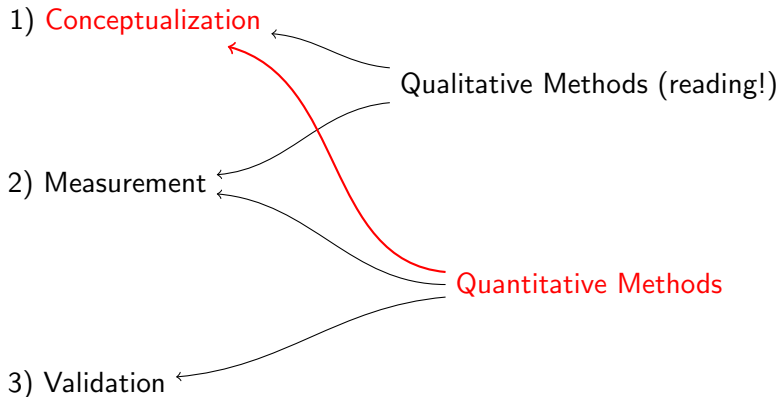


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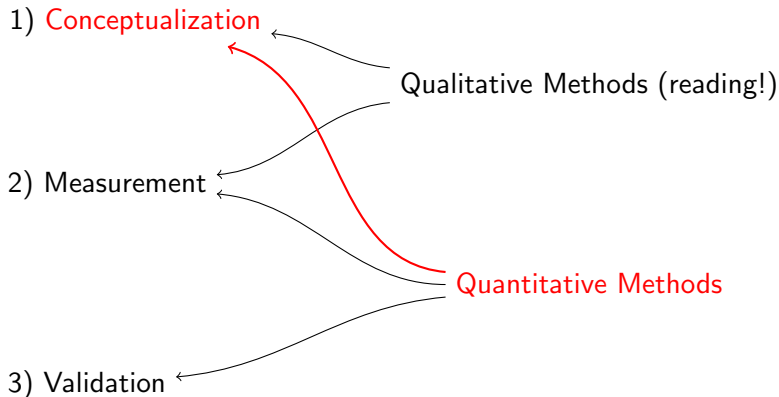


Advancing the Objective of Discovery



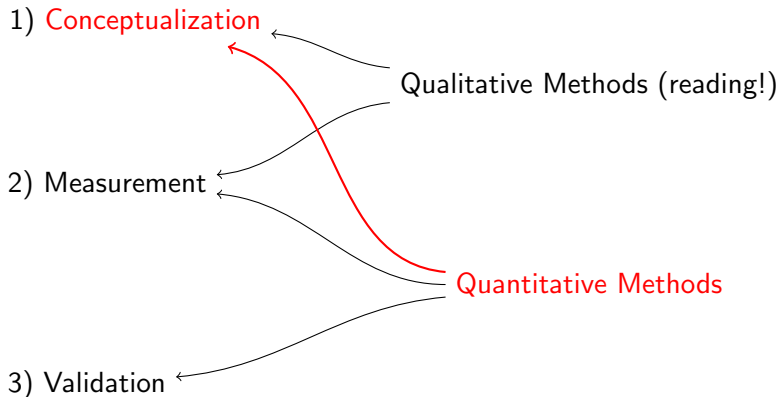
Quantitative methods for conceptualization: aiding **discovery**

Advancing the Objective of Discovery



- Quantitative methods for conceptualization: aiding **discovery**
- Few formal methods designed explicitly for conceptualization

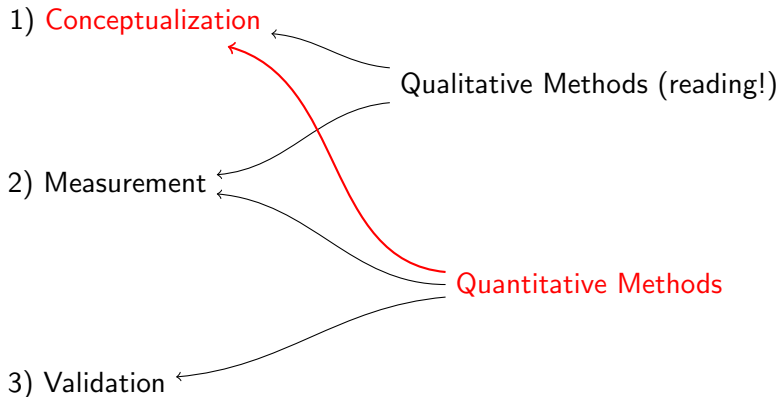
Advancing the Objective of Discovery



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Advancing the Objective of Discovery



Quantitative methods for conceptualization: aiding **discovery**

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- Evaluation methods measure progress in discovery

For more information

<http://GKing.Harvard.edu>